

Credit Card Transaction and Customer Analysis weekly

The objective of this project is to analyze credit card transaction data and customer information to identify revenue trends, customer behavior, spending patterns, transaction preferences, and business opportunities.

Two interactive Power BI dashboards were developed:

- Credit Card Transaction Report
- Customer Credit Card Report

The dashboards provide insights into transaction performance, customer demographics, revenue distribution, spending categories, card usage, activation rates, delinquency rates, and week-over-week revenue growth.

Project Objectives:

The main objectives of this project were:

- Analyze total revenue generated by customers.
- Study transaction patterns.
- Understand customer demographics.
- Identify high-value customers.
- Analyze card category performance.
- Evaluate spending behavior.
- Measure customer activation.
- Monitor delinquent customers.
- Track weekly revenue growth.
- Create interactive dashboards for business decision-making.

KPIs:

The dashboard highlights the following Key Performance Indicators (KPIs):

55 M	7.84 M	45 M	656 K
Total Revenue	Total Interest Earned	Total Transaction Amount	Total Transactions

1 – Credit Card Transaction Analysis

Insight 1: Quarterly Revenue Analysis

Revenue remained relatively stable throughout the year.

- Q1 generated approximately **14M**
- Q2 remained similar
- Q3 recorded the highest revenue
- Q4 experienced a slight decline

Observation

- Although transaction count peaked in Q3, revenue remained fairly consistent across all quarters.
- Business Insight:
- Higher transaction counts do not always lead to significantly higher revenue.
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Insight 2: Card Category Performance

<i>Card Type</i>	<i>Revenue</i>
<i>Blue</i>	46.14M
<i>Silver</i>	5.59M
<i>Gold</i>	2.45M
<i>Platinum</i>	1.14M

Observation

- Blue cards generated approximately 84% of total revenue.

Business Insight:

- Blue card holders represent the largest customer segment and contribute the majority of business revenue.

Insight 3: Interest Earned

Interest earned totaled

7.84 Million

This indicates a significant portion of revenue comes from interest payments in addition to transaction fees.

Insight 4: Revenue by Expenditure Type

Category	Revenue
Bills	13.8M
Entertainment	9.5M
Fuel	9.3M
Grocery	8.6M
Food	8.3M
Travel	5.9M

Observation

- Bills generated the highest revenue.
- Travel generated the lowest revenue.

Business Insight:

- Customers primarily use credit cards for essential expenses rather than travel.

Insight 5: Transaction Method

Revenue by transaction method:

Method	Revenue
Swipe	34.91M
Chip	16.97M
Online	3.44M

Observation:

- Swipe transactions dominate customer activity.
- Online transactions contribute only a small percentage.

Business Insight:

Customers still prefer physical card transactions over online payments.

Insight 6: Transaction Count by Card Usage

The waterfall chart shows:

- Swipe transactions contribute the largest transaction volume.
- Chip transactions are the second most popular.
- Online transactions remain comparatively low.

Business Insight:

Digital payment adoption can be improved through promotional campaigns.

2 – Customer Analysis weekly

Insight 7: Revenue by Gender

- Male Customers: 30.22M
- Female Customers: 25.09M

Observation:

Male customers contribute slightly more revenue than female customers.

Insight 8: Revenue Trend Over Time

The time series chart shows monthly fluctuations in revenue.

Observation:

Revenue remains relatively stable throughout the year with several seasonal peaks.

Business Insight:

Business performance is consistent without major declines.

Insight 9: Revenue by Age Group

Age Group Revenue

20–30	Lowest
30–40	Moderate
40–50	Highest
50–60	Second Highest
60+	Lowest

Observation:

Customers aged 40–50 generate the highest revenue.

Business Insight:

Middle-aged customers are the most valuable customer segment.

Insight 10: Revenue by Income Group

Revenue distribution:

- High Income: 22M+
- Medium Income: Approximately 16M
- Low Income: Approximately 10M

Observation:

High-income customers generate the largest share of revenue.

Business Insight:

Premium products should focus on high-income customers.

Insight 11: Revenue by Education Level

<i>Education</i>	<i>Revenue</i>
<i>Graduate</i>	22.30M
<i>High School</i>	11.12M
<i>Unknown</i>	8.48M
<i>Uneducated</i>	8.19M
<i>Post Graduate</i>	2.79M
<i>Doctorate</i>	2.42M

Observation:

Graduates generate the highest revenue.

Business Insight:

Marketing campaigns targeted toward graduates may yield higher returns.

Insight 12: Revenue by State

Top contributing states include:

- Texas
- New York
- California

Observation:

These states generate the majority of customer revenue.

Business Insight:

Regional marketing campaigns should prioritize these high-performing states.

3 – Weekly Performance Report

Insight 13: Week-over-Week Revenue Growth

The weekly report compares:

- Previous Week Revenue
- Current Week Revenue
- Weekly Growth Percentage

Examples:

- Week 3 Revenue increased by 9.0%
- Week 4 Revenue decreased by 6.6%
- Week 11 Revenue increased by 14.5%

Business Insight:

Weekly monitoring enables businesses to quickly identify positive or negative revenue trends.

Insight 14: Customer Delinquency Rate

Delinquent Customers: 6.07%

Non-Delinquent Customers: 93.93%

Observation:

Only a small percentage of customers are delinquent.

Business Insight:

Overall customer credit health appears stable.

Insight 15: Card Activation Rate

Activated Cards: 57.47%

Inactive Cards: 42.53%

Observation:

Nearly half of customers have not activated their cards.

Business Insight:

Banks can improve revenue by encouraging inactive customers to activate and use their cards.

Overall Business Insights

The analysis revealed several important findings:

- Blue credit cards generate the majority of company revenue.
- Swipe transactions remain the preferred payment method.
- Bills represent the highest spending category.
- Customers aged 40–50 are the most profitable segment.
- High-income customers contribute significantly more revenue than other income groups.
- Graduates generate the highest revenue among education levels.
- Male customers slightly outperform female customers in terms of revenue.
- Texas, New York, and California are the strongest-performing regions.
- Card activation rates indicate an opportunity to engage inactive users.
- Delinquency rates are relatively low, suggesting a healthy customer portfolio.
- Weekly revenue tracking helps identify growth opportunities and periods of decline.

Business Recommendations

- Promote premium credit card offers to Blue card customers to encourage upgrades.
- Increase incentives for online transactions to improve digital payment adoption.
- Target customers aged 40–50 with personalized marketing campaigns.
- Develop loyalty programs for high-income customers to enhance retention.
- Encourage inactive customers to activate their cards through promotional offers.
- Focus marketing efforts on high-performing states such as Texas, New York, and California.
- Offer cashback and rewards on travel-related spending to boost revenue in the lowest-performing category.
- Continuously monitor weekly revenue trends to respond quickly to changes in customer behavior.